Demo Smart Plug: Quick-start Integration Guide

This guide is for developers who integrate the device with their own application. It describes the data points the device exposes and the generic command/report pattern. Map the message shapes below onto the protocol of your IoT platform; the shapes here are illustrative and not tied to any vendor.

1. Before you start

- A provisioned device of category Smart plug using Wi-Fi.
- Credentials for your IoT platform. Keep them in your platform's secret store, never in source code.
- The thing model file thing-model.json shipped with this guide.

2. Provisioning

Supported provisioning methods:

- Hotspot (SoftAP)
- BLE-assisted

3. Data points

ID is the primary key. Use the ID when you send commands and when you parse reports.

Control	Identifier	ID	Access	Interaction
Power switch	power_switch	1	RW	Tap to toggle; command/report loop
Active power	active_power	2	R	Read-only number with a small trend chart
Overload alarm	overload_alarm	3	R	Read-only; shown in red while the alarm is active
Overload protection	overload_protect	4	RW	Slide switch; command/report loop
Timer tasks	timer_task	5	RW	List with add/edit/delete; wait for device report after each command
Report interval	report_interval	6	RW	Slider in minutes; command/report loop

Data types and ranges:

- ID 1 power_switch (Power switch): bool (true/false)
- ID 2 active_power (Active power): float, range 0 to 99999, step 0.1, unit W
- ID 3 overload_alarm (Overload alarm): bool (true/false)
- ID 4 overload_protect (Overload protection): bool (true/false)
- ID 5 timer_task (Timer tasks): string, max length 1024
- ID 6 report_interval (Report interval): int, range 1 to 60, step 1, unit min

4. Reading values (report)

The device reports properties with access R or RW. Illustrative report body:

```
{
  "1": true,
  "2": 0,
  "3": true
}
```

5. Changing values (command)

Only properties with access RW or W accept commands. Illustrative command body:

```
{
  "1": true
}
```

After a command, wait for the next report of the same ID before you treat the change as applied.

6. Offline behaviour

Panel greys out writable controls and shows the last reported values with their timestamp.

7. Troubleshooting

- A command has no effect: check that the property access is RW or W and that the value is inside the range.
- Values look stale: check the report interval property, if the product has one, and the device connection.
- IDs do not match: regenerate this guide from thing-model.json and run the consistency check.